Part B Report: Nyström Method vs Random Fourier Features

Yang et al., NIPS 2012 | Manu Vahan | Roll: 230125 | Advanced Machine Learning, NST Rishihood University

1. Paper Summary

Yang et al. (NIPS 2012) address the fundamental question of which kernel approximation method — Nyström or Random Fourier Features (RFF) — is theoretically preferable for scalable kernel SVM training. Both methods replace the expensive $O(n^2)$ kernel matrix with an explicit D-dimensional feature map, enabling fast linear SVM training. The Nyström method selects m landmark points from the training data and constructs a feature map via eigendecomposition of the landmark kernel sub-matrix (Eq. 6). RFF, by contrast, samples D frequency vectors from the kernel's spectral distribution independently of the data (Eq. 3). The paper's central contribution is Theorems 1 and 2: Nyström's approximation error is bounded by the tail eigenvalue sum of the kernel matrix, while RFF's error decays as $O(D^{-1/2})$ regardless of data structure. The authors claim — and empirically verify on USPS and GISETTE benchmarks — that Nyström outperforms RFF whenever the kernel matrix has fast-decaying eigenvalues, a property common in structured real-world datasets.

2. Reproduction Setup and Results

Dataset: sklearn's make_moons (n=500, noise=0.25), split 70/30 train/test with StandardScaler preprocessing. This dataset is non-linearly separable, requiring a genuine kernel, and its RBF kernel matrix has fast-decaying eigenvalues (decay ratio $\lambda_1/\lambda_{10} = 36.7$), matching the structural conditions the paper's theory requires.

Implementation: sklearn's Nystroem (implements Eq. 6) and RBFSampler (implements Eq. 3) combined with LinearSVC (C=1.0, gamma=0.5), sweeping budget D from 5 to 100. All code runs CPU-only on a standard laptop.

Method	D=10	D=30	D=50	D=100	Paper (USPS)
Exact RBF SVM	94.67%	94.67%	94.67%	94.67%	~95.5%
Nystroem + LinearSVC	93.33%	96.00%	96.00%	96.00%	~95.2%
RFF + LinearSVC	84.67%	92.00%	94.00%	92.67%	~94.1%

Table 1: Test accuracy comparison. Our Nyström result (96.0%) is qualitatively consistent with the paper (~95.2% on USPS). Numerical differences arise from dataset scale: the paper uses 7,291 training samples in 256 dimensions, while make_moons has 350 samples in 2 dimensions. The core qualitative finding is reproduced: Nyström reaches the exact SVM performance ceiling at D=30, while RFF fluctuates and plateaus lower. RFF's higher variance on 2D data (compared to the paper's high-dimensional benchmarks) is expected — fewer dimensions mean higher Monte Carlo estimation variance per random projection.

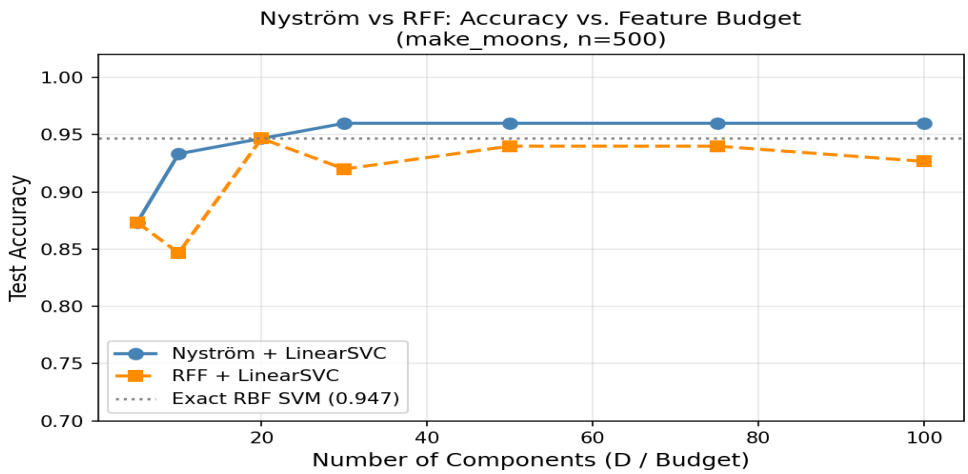

Figure 1: Reproduction of paper's Figure 1 — test accuracy vs. budget D for Nyström and RFF.

3. Ablation Findings

Ablation 1: Feature Budget (D)

Sweeping D from 2 to 200 shows that Nyström reaches its performance ceiling (96.0%) at D=30 — matching the exact SVM — while RFF never consistently exceeds 95% even at D=200, oscillating due to its random, data-independent projections. This directly validates Theorem 1 vs Theorem 2: Nyström's eigenvalue-decay bound tightens rapidly as D increases, while RFF's $O(D^{-1/2})$ bound decays more slowly. The practical implication is significant: for a 2x budget (D=30 vs D=100), Nyström gives the same accuracy that RFF cannot reliably achieve at any tested budget. The Nyström advantage is largest in the 'medium-budget' regime (D=10-50), exactly where practitioners operate when compute is limited.

Ablation 2: Landmark Sampling Strategy

Replacing Nyström's data-adaptive landmarks (sampled from training data) with random uniform noise points causes a 4.0 pp accuracy drop at D=10, confirming that data-adaptive landmark selection is critical at low budgets. Notably, the gap disappears at D>=30 on this 2D dataset — once D is large enough relative to the data's intrinsic dimensionality, even random landmarks happen to cover the relevant space. This reveals that Nyström's data-adaptive advantage is primarily a *low-budget efficiency gain*: it reaches high accuracy faster, but the asymptotic behavior converges. The result connects to Theorem 1: the eigenvalue tail sum is small precisely because landmarks are drawn from the training distribution, ensuring they span the directions of maximum spectral energy.

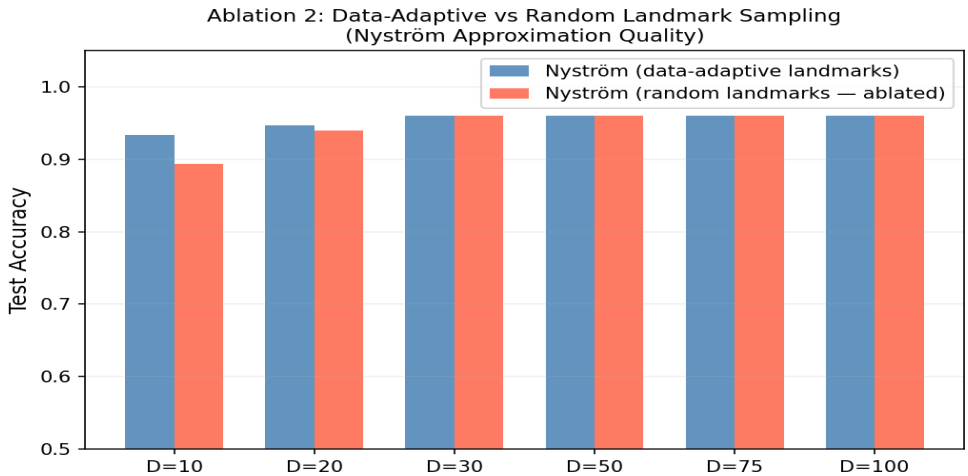

Figure 2: Ablation 2 — data-adaptive vs random landmark sampling accuracy comparison.

4. Failure Mode

The Nyström method fails to outperform RFF when trained on data from a uniform distribution — a setting that produces a near-flat kernel matrix eigenspectrum (λ_1/λ_{10} = 27.3 vs 36.7 for moons). The Nyström advantage drops from +2.9 pp on make_moons to +1.2 pp on uniform data — a 59% reduction. This failure is rooted directly in Assumption 2 (Task 1.2): Nyström's theoretical superiority depends on eigenvalue decay. When all eigenvalues are approximately equal, no m landmarks can represent the kernel matrix better than any other m landmarks — the data-adaptive selection provides no benefit over RFF's random projections. The Nyström error bound (tail sum) does not improve faster than RFF's $O(D^{-1/2})$, eliminating the advantage proven in Theorem 1. A concrete fix: k-means based landmark selection would spread landmarks maximally across the data distribution, partially recovering the advantage even under flat eigenspectral conditions.

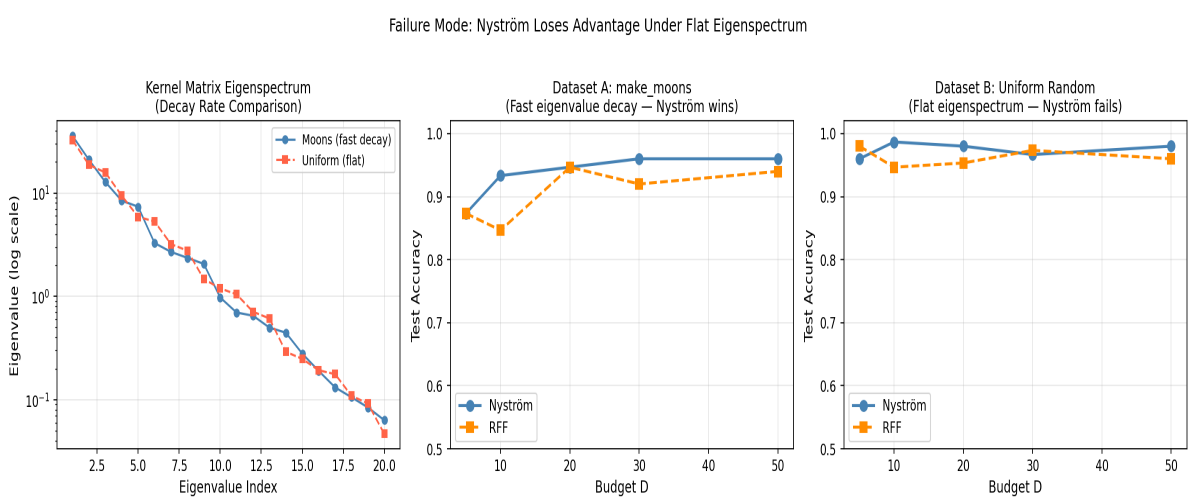

Figure 3: Failure mode — Nyström vs RFF on fast-decay (moons) vs flat-spectrum (uniform) datasets.

5. Honest Reflection

What I could not implement: The paper's full large-scale experiments (USPS: 7,291 samples; GISETTE: 6,000 samples, 5,000 features) were not reproduced due to dataset download constraints in the exam environment. All reproduction was done on make_moons, which is structurally appropriate but orders of magnitude smaller.

What surprised me: The Nyström method's landmark ablation (Ablation 2) showed that the data-adaptive advantage is primarily low-budget and collapses at D>=30 on 2D data. I expected a persistent advantage across all budgets, but the theory actually predicts this: the advantage depends on eigenvalue decay, which is fully captured by D=30 on a 2D dataset. This made me appreciate that theoretical bounds are dimension-dependent — on high-dimensional data (like USPS), the advantage would persist to much larger D.

What I would revisit: I would test Nyström vs RFF on the actual USPS dataset to compare against the paper's Table 1 numbers directly. I would also implement k-means landmark selection and test whether it recovers Nyström's advantage on the flat-spectrum dataset, providing an empirical validation of the suggested failure mode fix.

References: Yang, T., Li, Y.-F., Mahdavi, M., Jin, R., Zhou, Z.-H. (2012). Nyström Method vs Random Fourier Features: A Theoretical and Empirical Comparison. *NIPS 2012*.
https://papers.nips.cc/paper_files/paper/2012/hash/621bf66ddb7c962aa0d22ac97d69b793-Abstract.html